

Lesson: Public vs Private API Access (X vs GitHub)

Introduction

This lesson explains why some platforms require login for API access while others allow public access. It helps students understand data privacy, API authentication, and responsible data usage.

1. Understanding the Platforms

- **X (formerly Twitter):** Social media platform where posts may contain sensitive personal data.
- **GitHub:** Code hosting platform where most projects are public and open to collaboration.

2. Public vs Private Data

- **X:** Personal posts, conversations, private accounts → requires login and authentication
- **GitHub:** Public repositories, open-source code → no login needed for basic access.

3. API Access and Authentication

X API:

- Requires authentication (Bearer Token, OAuth) for private data.
- Protects users and enforces rate limits.

GitHub API:

- Public data can be accessed without login.
- Authentication is optional, mainly for higher rate limits or private repositories.

4. Learning

1. Use analogies: X = Bank (private), GitHub = Public Library (open).

2. Show live examples:
 - GitHub API without token → success
 - X API without token → Unauthorized error
3. Discuss consequences of unauthorized scraping.
4. Hands-on mini-lab: students practice GitHub API.
5. Reflection: importance of privacy and responsible data usage.

Although there is no hands-on with X API, the workflow is as below for understanding:

- Create a developer account on the X Developer Portal
- Create an application/project
- Generate a Bearer Token
- Use the token in API requests

5. Practical Mini Lab (GitHub API)

Example: Fetch public repositories of a user without login

```
import requests

url = 'https://api.github.com/users/octocat/repos'
response = requests.get(url)
data = response.json()

print('Total repos:', len(data))
print('First repo name:', data[0]['name'])
```

6. Discussion Questions (In-Class)

- Why does X require login while GitHub does not?
- How does authentication protect data?
- What are ethical considerations when collecting API data?
- How do rate limits affect your program?